# EECS 570
# Lecture 8
# Bus-based SMPs


**Fall 2025**

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**http://www.eecs.umich.edu/courses/eecs570/**

Slides developed in part by Profs. Adve, Falsafi, Hill, Lebeck, Martin, Narayanasamy, Nowatzyk, Reinhardt, Roth, Smith, Singh, and Wenisch.

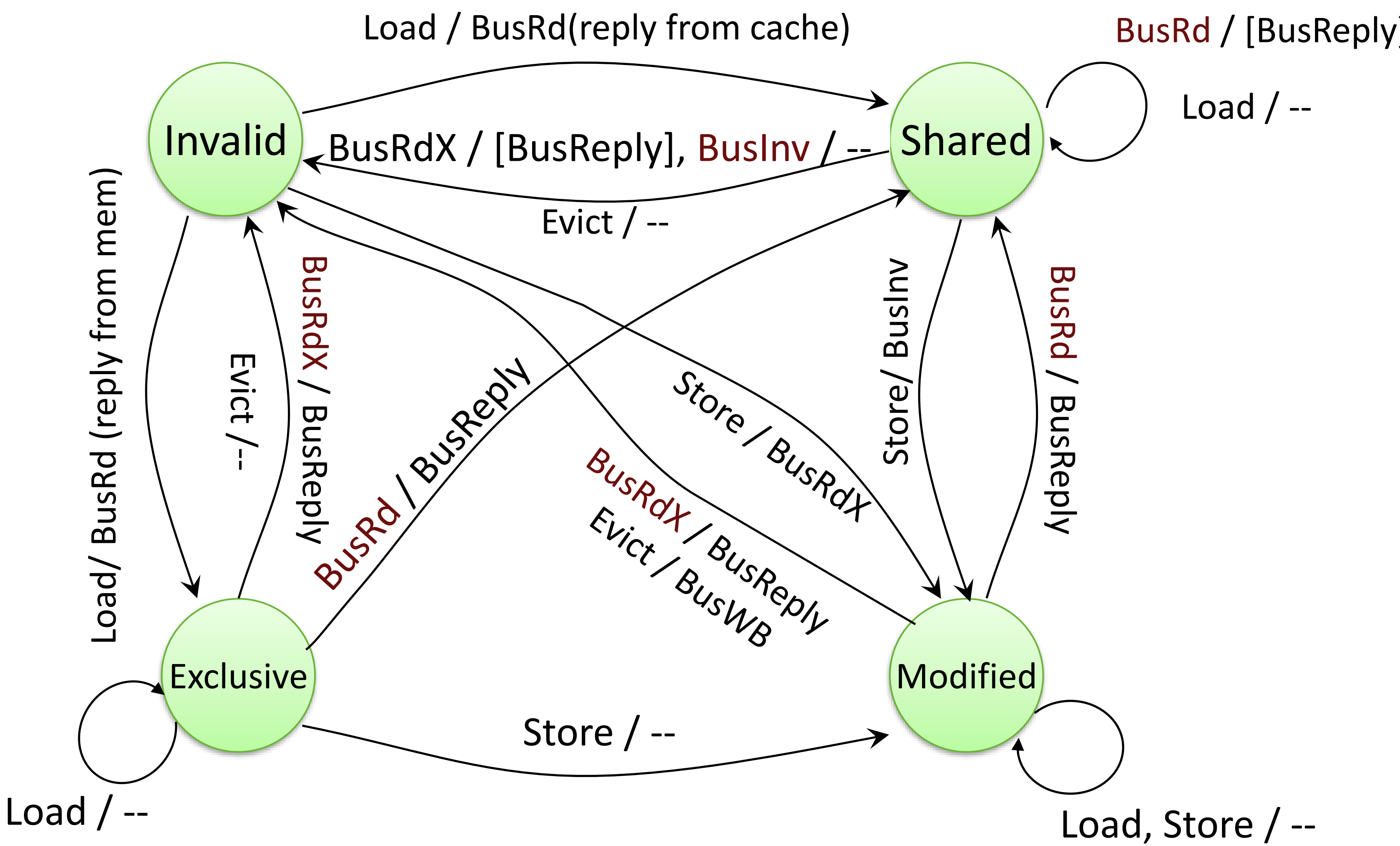
# Reading this week

Daniel J. Sorin, Mark D. Hill, and David A. Wood, A Primer on Memory Consistency and Cache Coherence (Ch. 6 & 7)

# MESI Protocol (aka Illinois)

- MSI suffers from frequent read-upgrade sequences
  - Leads to two bus transactions, even for private blocks
  - Uniprocessors don’t have this problem
- Solution: add an “Exclusive” state
  - Exclusive – only one copy; writable; clean
    - Can detect exclusivity when memory provides reply to a read
  - Stores transition to Modified to indicate data is dirty
    - No need for a BusWB from Exclusive

# MESI Protocol Summary



# MOESI Protocol

- MESI must write-back to memory on M→S transitions
  - Because protocol allows "silent" evicts from shared state, a dirty block might otherwise be lost
  - But, the writebacks might be a waste of bandwidth
    - E.g., if there is a subsequent store
    - Common case in producer-consumer scenarios

- Solution: add an "Owned" state
  - Owned – shared, but dirty; only one owner (others enter S)
    - Entered on M→S transition, aka "downgrade"
  - Owner is responsible for writeback upon eviction

# MOESI Framework

[Sweazey & Smith ISCA86]

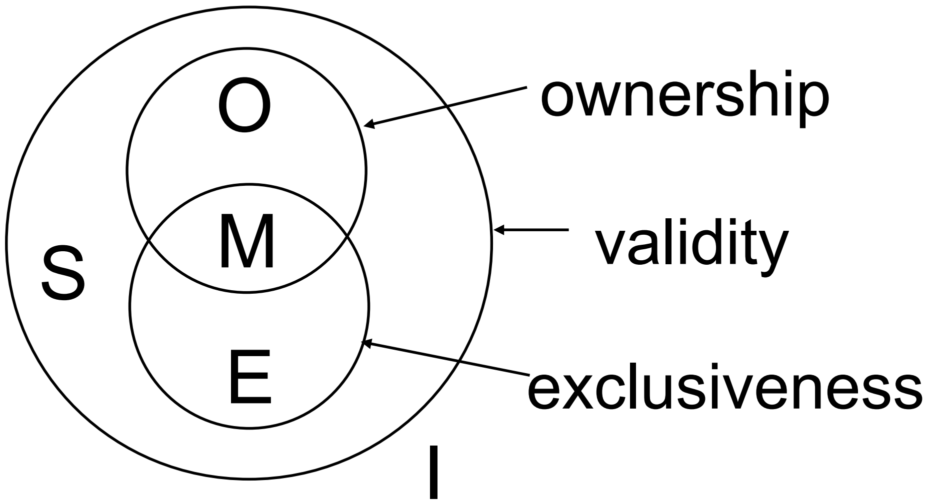
M - Modified (dirty)

O - Owned (dirty but shared) WHY?

E - Exclusive (clean unshared) only copy, not dirty

S - Shared

I - Invalid



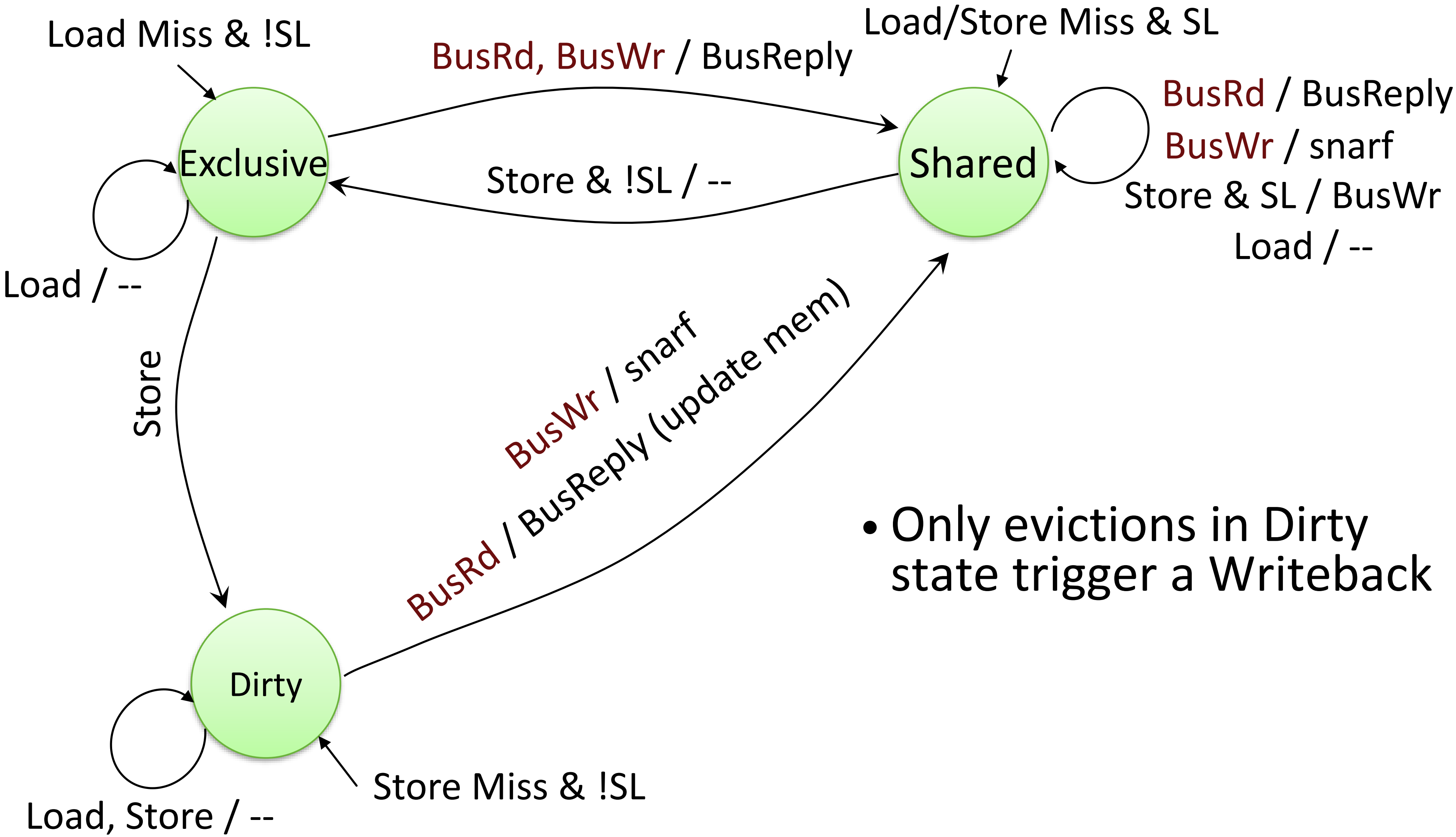

Variants

- MSI
- MESI
- MOSI
- MOESI

# DEC Firefly

- An update protocol for write-back caches
- States
  - Exclusive – only one copy; writable; clean
  - Shared – multiple copies; write hits write-through to all sharers and memory
  - Dirty – only one copy; writeable; dirty
- Exclusive/dirty provide write-back semantics for private data
- Shared state provides update semantics for shared data
  - Uses “shared line” bus wire to detect sharing status
- Well suited to producer-consumer; process migration hurts

# DEC Firefly Protocol Summary




- Only evictions in Dirty state trigger a Writeback

# Non-Atomic State Transitions

Operations involve multiple actions

- Look up cache tags
- Bus arbitration
- Check for writeback
- Even if bus is atomic, overall set of actions is not
- Race conditions among multiple operations

Suppose P1 and P2 attempt to write cached block A

- Each decides to issue BusUpgr to allow S –> M

Issues

- Handle requests for other blocks while waiting to acquire bus
- Must handle requests for this block A

You'll see a lot of this in PA2! ☺

# Scalability problems of Snoopy Coherence

- Prohibitive **bus bandwidth**
  - Required bandwidth grows with # CPUS…
  - … but available BW per bus is fixed
  - Adding busses makes serialization/ordering hard
- Prohibitive **processor snooping bandwidth**
  - All caches do tag lookup when ANY processor accesses memory
  - Inclusion limits this to L2, but still lots of lookups

- **Upshot**: bus-based coherence doesn't scale beyond 8–16 CPUs

# Implementing Snoopy Coherent SMPs

# Outline

- Coherence Control Implementation
- Writebacks, non-atomicity, serialization/order
- Hierarchical caches
- Split Busses
- Deadlock, livelock & starvation
- TLB Coherence

# Base Coherence SMP design

- Single-level write-back cache
- MSI coherence protocol
- One outstanding memory request per CPU
- Atomic memory bus transactions
  - No interleaving of transactions
- Atomic operations within process
  - One operation at a time in program order

- We will incrementally add more concurrency/complexity

# Cache Controller & Tags

- On a miss in a uniprocessor
  - Assert request for bus
  - Wait for bus grant
  - Drive address & command lines
  - Wait for command to be accepted by target device
  - Transfer data
- In a Snoop-based SMP, cache controller must:
  - Monitor bus and CPU
    - Can view as two controllers, bus-side and CPU-side
    - With a single cache level, tags often duplicated or dual-ported
  - Respond to bus transactions as needed

# Reporting Snoop results: How?

- Collective response from caches must appear on bus
- Wired-OR signals (used in Firefly protocol)
  - Shared: assert if any cache has a copy
  - Dirty/Inhibit: asserted if some cache has a dirty copy
    - Needn't indicate which; it knows what it needs to do
    - Also indicates that memory controller should ignore request
  - Snoop-valid: asserted when OK to check other two signals
- Need arbitration/priority scheme for cache-to-cache xfers
  - Which cache should supply data in shared state?

# Reporting Snoop results: When?

- Memory needs to know what, if anything, to do

- Solution 1: Fixed # of clocks after request message
  - Usually needs duplicate tags to avoid contention w/ CPU
  - Pentium Pro, HP Servers, Sun Enterprise

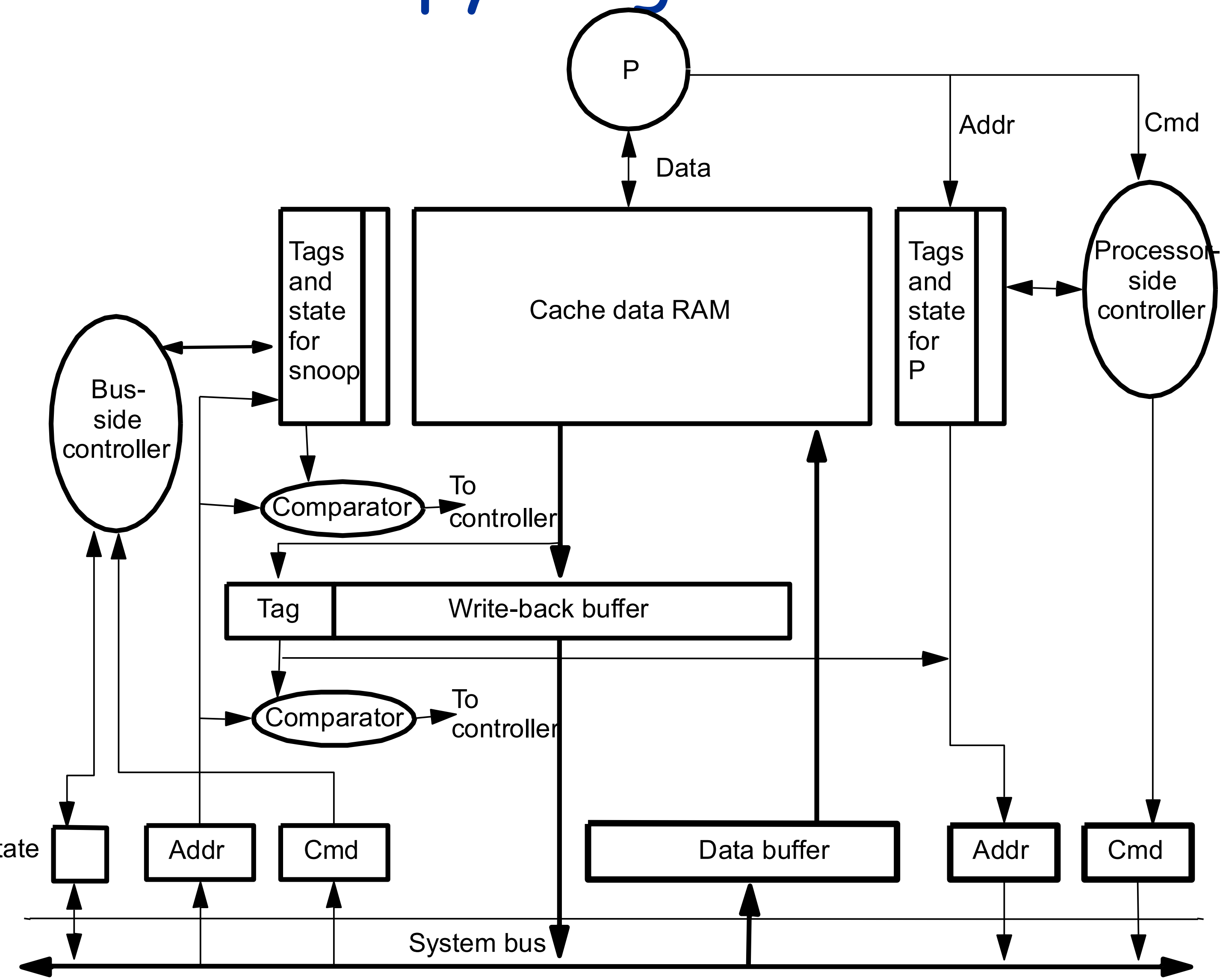
- Solution 2: Variable delay
  - Memory assumes cache will supply data until all say "sorry"
  - Less conservative, more flexible, more complex

# Writebacks

- Allow CPU to proceed on a miss ASAP
  - Fetch the requested block
  - Do the writeback of the victim later

- Requires write buffer
  - Must snoop/handle bus transactions in write buffer
  - Must maintain order of writes/reads (maintain consistency)

# Base Snoopy Organization



# Serialization and Ordering

- CPU-cache handshake must preserve serialization
  - E.g., write in S state → first obtain permission

- Write completion for sequential consistency (SC)

→ need to send invalidations
  - Wait to get bus, then can consider writes complete

  - Must serialize bus transactions in program order
    - Split transaction bus still must retire transactions in order

# The Inclusion Property

- Inclusion means L2 is a superset of L1 (ditto for L3…)
  - Also, must propagate “dirty” bit through cache hierarchy
- Now, only need to snoop last level cache
  - If L2 says not present, can’t be in L1 either
- **Inclusion takes effort to maintain**
  - L2 must track what is cached in L1
  - On L2 replacement, must flush corresponding blocks from L1

    *How can this happen?*

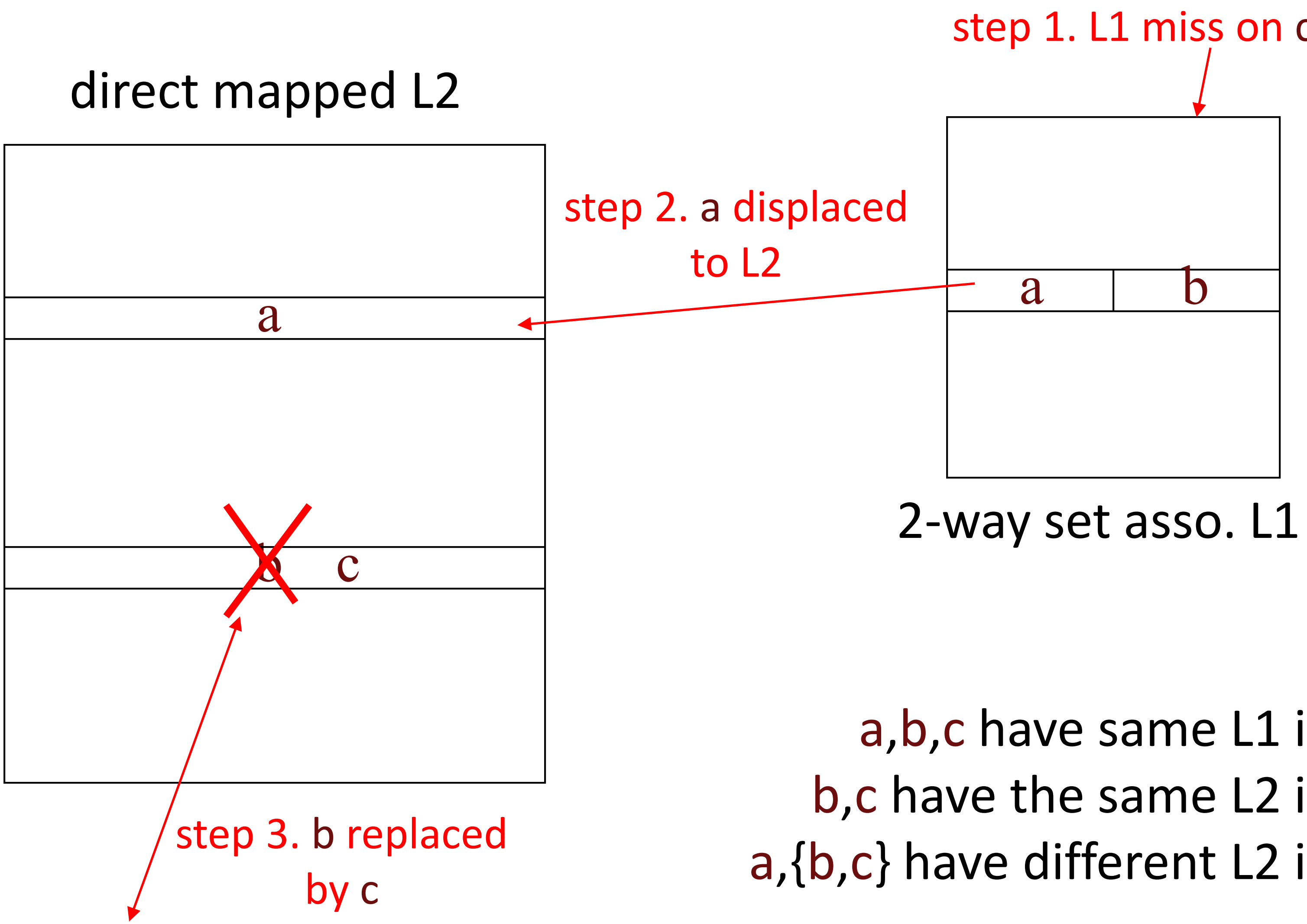
    *Consider:*

    *1. L1 block size < L2 block size*

    *2. different associativity in L1*

    *3. L1 filters L2 access sequence; affects LRU ordering*

# Possible Inclusion Violation




a,b,c have same L1 idx bits

b,c have the same L2 idx bits

a,{b,c} have different L2 idx bits

# Multi-level Cache Hierarchies

- How to snoop with multi-level caches?
  - Independent bus snooping at each level?
  - Multiple duplicate tag arrays
  - Maintain cache **inclusion**

# Is inclusion a good idea?

- Most common inclusion solution:
  - Ensure L2 holds a superset of L1I and L1D
  - On L2 replacement or coherence action that supplies data, forward actions to L1s
- But…
  - Restricted L2 associativity may limit blocks in split L1s
  - Not that hard to always snoop the L1s
- Many recent designs do not maintain inclusion
  - Leads to more complex coherence protocols

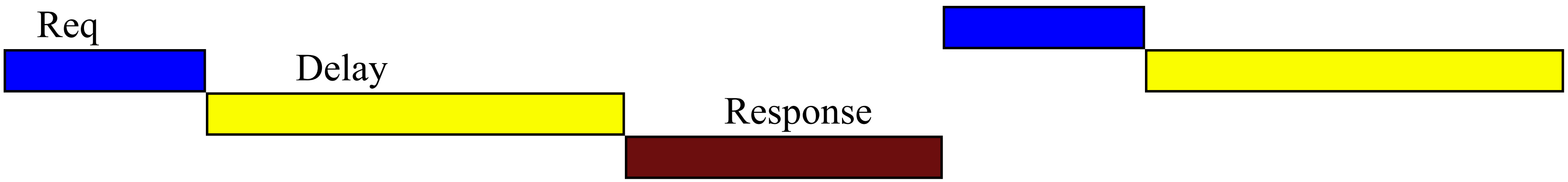
# Shared Caches

- Share low level caches among multiple processors
  - Sharing L1 adds to latency, *unless* multithreaded processor
- Advantages
  - Eliminates need for coherence protocol at shared level
  - Reduces latency within sharing group
  - Processors essentially prefetch for each other
  - Can exploit working set sharing
  - Increases utilization of cache hardware
- Disadvantages
  - Higher bandwidth requirements
  - Increased hit latency
  - May be more complex design
  - Lower effective capacity if working sets don't overlap

# Split-transaction (Pipelined) Bus

- Supports multiple simultaneous transactions

Atomic Transaction Bus




Split-transaction Bus

# Potential Problems

- Two transactions to same block (conflicting)
  - Mid-transaction snoop hits
- Buffer requests and responses
  - Need flow control to prevent deadlock
- Ordering of Snoop responses
  - when does snoop response appear wrt data response

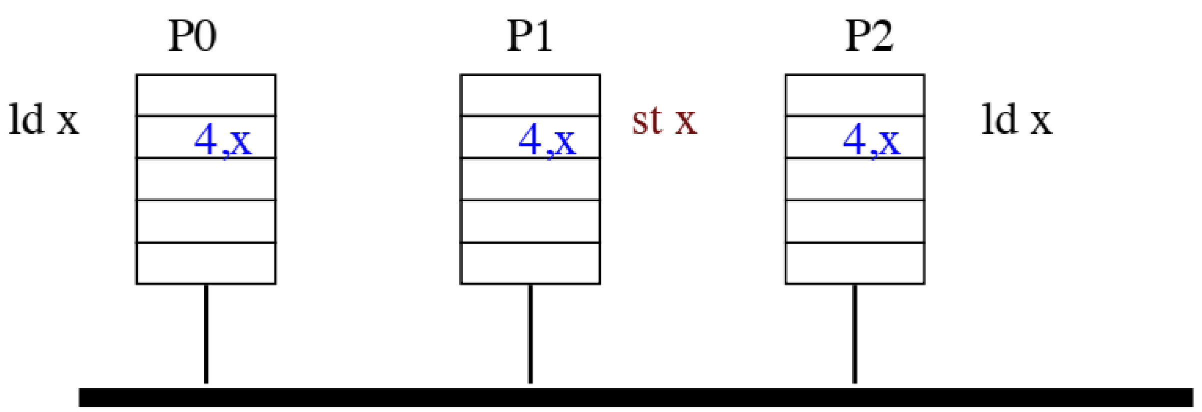
# Possible Solutions

- Disallow conflicting transactions
- NACK for flow control
- Out-of-order responses
  - snoop results presented with data response

# Case Study: Sun Enterprise 10000

- How far can you go with snooping coherence?
- Quadruple request/snoop bandwidth using four address busses
  - each handles 1/4 of physical address space
  - impose *logical* ordering: for writes on same cycle, those on bus 0 occur "before" bus 1, etc.
- Get rid of data bandwidth problem: use a network
  - E10000 uses 16x16 crossbar betw. CPU boards & memory boards
  - Each CPU board has up to 4 CPUs: max 64 CPUs total
- 10.7 GB/s max BW, 468 ns unloaded miss latency
- See "Starfire: Extending the SMP Envelope", IEEE Micro 1998

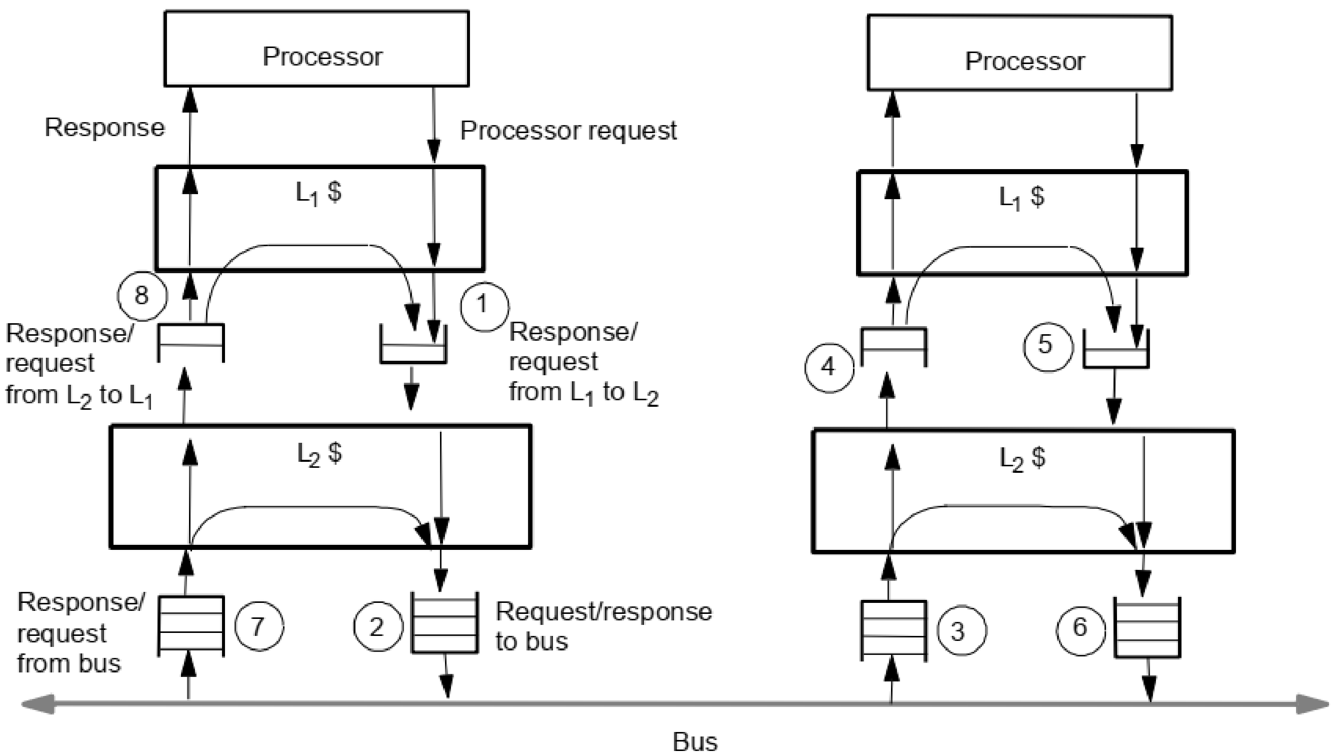
# Split-Transaction Bus Example




Per-processor request table tracks all transactions

P2 Can snoop data from first ld
P1 Must hold st operation until entry is clear

# Multi-Level Caches with Split Bus

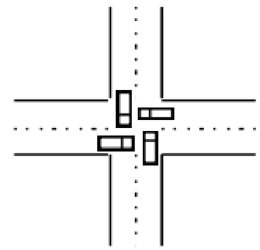


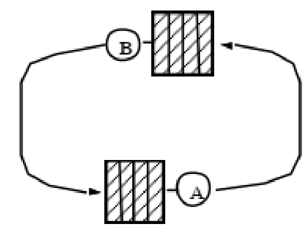
# Multi-level Caches with Split-Transaction Bus

- General structure uses queues between
  - Bus and L2 cache
  - L2 cache and L1 cache
- Deadlock!
- Classify all transactions
  - Request, only generates responses
  - Response, doesn't generate any other transactions
- Requestor guarantees space for all responses
- Use Separate Request and Response queues
  - This ideal will evolve into "virtual channels" in Unit 3

# More on Correctness

- Partial correctness (never wrong): Maintain coherence and consistency (safety)
- Full correctness (always right): Prevent: (liveness)
- Deadlock:
  - all system activity ceases
  - Cycle of resource dependences






- Livelock:
  - no processor makes forward progress
  - constant on-going transactions at hardware level
  - e.g. simultaneous writes in invalidation-based protocol
- Starvation:
  - some processors make no forward progress
  - e.g. interleaved memory system with NACK on bank busy

# Deadlock, Livelock, Starvation

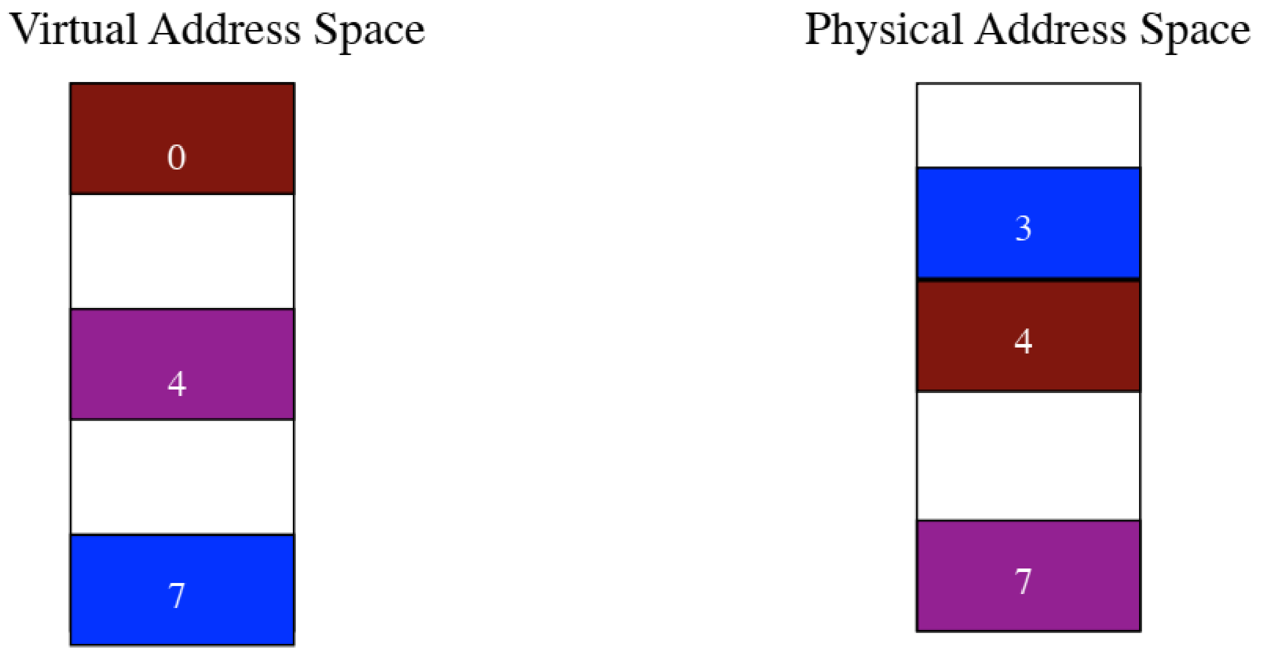
- Request-reply protocols can lead to *deadlock*
  - When issuing requests, must service incoming transactions
  - e.g. cache awaiting bus grant must snoop & flush blocks
  - else may not respond to request that will release bus: deadlock
- Livelock:
  - window of vulnerability problem [Kubi et al., MIT]
  - Handling invalidations between obtaining ownership & write
  - Solution: don't let exclusive ownership be stolen before write *
- Starvation:
  - solve by using fair arbitration on bus and FIFO buffers

# Deadlock Avoidance

- Responses are never delayed by requests waiting for a response
- Responses are guaranteed to be sunk
- Requests will eventually be serviced since the number of responses is bounded by outstanding requests
- Must classify transactions according to deadlock and coherence semantics

# Translation Lookaside Buffer

- Cache of Page Table Entries
- Page Table Maps Virtual Page to Physical Frame



# The TLB Coherence Problem

- Since TLB is a cache, must be kept coherent
- Change of PTE on one processor must be seen by all processors
- Process migration
- Changes are infrequent
  - get OS to do it
  - Always flush TLB is often adequate

# TLB Shootdown

- To modify TLB entry, modifying processor must
  - LOCK page table,
  - flush TLB entries,
  - queue TLB operations,
  - send interprocessor interrupt,
  - spin until other processors are done
  - UNLOCK page table

- SLOW…
  - But most common solution today
- Some ISAs have "flush TLB entry" instructions